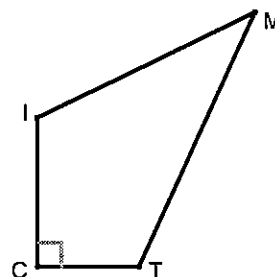


- Let k be the smallest positive solution for x for the equation $\sin x = 0.6$ where x is measured in degrees. Let w be the smallest positive solution for y for the equation $\sin y = 0.6$ where y is measured in radians. Determine the product (kw) . Express your answer as a decimal rounded to four significant digits.
- The minimum value of $f(x) = A \sin(2x + \pi) + 3$, where $A > 0$, is -7 . Determine the value of A .
- $\tan \theta = -\frac{7}{4}$, $\frac{\pi}{2} < \theta < \frac{3}{2}\pi$. Then $\sin \theta = \frac{k\sqrt{w}}{p}$ in reduced form with k , w , and p integers and $p > 0$. Determine the sum $(k + w + p)$.
- $\begin{cases} x = 4t + 5 \\ y = 5t - 1 \end{cases}$. Determine the slope of the line perpendicular to the graph of this parametric function when $t = -2$.
- For all permissible values of θ , $\frac{10 \sin^{(4)}(\theta) - 17 \sin^{(2)}(\theta) + 6}{18 - 15 \sin^{(2)}(\theta)} = k(\cos(2\theta))$. Determine the value of k . Express your answer as an integer or as a common or improper fraction.
- In quadrilateral $ICTM$, $\angle ICT = 90^\circ$ and $\angle CTM = 115^\circ$. Also, $IC = 8$, $CT = 6$ and $TM = 11$. Determine the numeric area of quadrilateral $ICTM$. Express your answer as a decimal rounded to the nearest thousandths.



- $f(x) = \frac{ax^2 - 11x + 5}{4 - 11x + 6x^2}$ has a horizontal asymptote at $f(x) = \frac{1}{3}$. Also, b represents the number of distinct vertical asymptotes for the graph of $f(x)$ and c represents the greatest number that is not an element of the range of $f(x)$. Determine the value of $f(b) + c$. Express your answer as an integer or as a common or improper fraction.